

WEEK 3

Task 1 — Machine Learning Fundamentals

Introduction: What Is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from data and use those patterns to make predictions or decisions, rather than relying only on rules written by a programmer. An ML system is trained using examples. After training, it can make predictions on new data, although its accuracy depends on the quality and amount of data and on how well the model fits the problem.

Example: An email service can learn from previously labelled messages to identify likely spam. Instead of checking only a fixed list of suspicious words, the model can use many patterns in message content and sender behaviour.

AI vs. Machine Learning vs. Deep Learning

Artificial Intelligence is the broad field of building systems that perform tasks associated with human intelligence, such as reasoning, planning, perception, and language understanding. Machine Learning is one approach within AI: systems improve their performance by learning from data. Deep Learning is a subfield of ML that uses neural networks with many layers to learn complex patterns, often from large datasets.

Relationship: AI is the broadest category; ML is part of AI; and deep learning is part of ML. For example, a rule-based chess program can be AI without using ML, a model that predicts house prices can use ML, and a neural network that recognises objects in photographs can use deep learning.

Dataset, Features, and Labels

A dataset is an organised collection of examples used to explore a problem or train and evaluate a model. It may be stored in a spreadsheet, CSV file, database, image collection, or other format. Each row often represents one example, and each column records a property of that example.

Features are the input information used by a model to make a prediction. Labels are the correct outcomes supplied for examples in supervised learning. In a house-price dataset, features could include floor area, number of bedrooms, and location; the label would be the selling price. In a spam dataset, features might describe the message and the label could be “spam” or “not spam.”

Training Data and Testing Data

The training dataset is used by an algorithm to learn patterns and fit a model. The testing dataset is kept separate and used after training to estimate how well the model performs on examples it has not seen before. A common split is 80% for training and 20% for testing, though the best choice depends on dataset size and the task. Data should be split carefully to prevent information from the test set leaking into training.

A model may memorise training examples instead of learning general patterns. This is called overfitting. Testing on unseen data helps reveal whether the model is likely to work on new examples.

Supervised and Unsupervised Learning

Supervised learning uses labelled examples: the dataset includes both inputs and known correct outputs. The model learns a relationship between the features and labels, then predicts labels for new inputs. Common tasks include classification and regression. Examples include predicting whether a transaction is fraudulent and estimating the price of a house.

Unsupervised learning works with data that has no supplied target labels. It searches for structure or patterns, such as groups of similar examples or compact representations of data. Clustering is a common technique. For example, a shop may group customers by purchasing behaviour to better understand different customer segments.

Classification vs. Regression

Classification predicts a category or class. Examples include identifying an email as spam or not spam, classifying a photo as a cat or dog, and predicting whether a customer may cancel a subscription.

Regression predicts a numerical value. Examples include estimating house prices, forecasting daily electricity demand, or predicting delivery time.

Quick comparison: Classification → a label (for example, “fraud” or “not fraud”). Regression → a number (for example, an estimated house price).

Real-World Applications of Machine Learning

- Healthcare: supporting analysis of medical images and helping estimate patient risk.
- Banking and finance: flagging unusual transactions and helping assess credit risk.
- Transport: estimating arrival times, forecasting traffic, and supporting driver-assistance systems.
- Retail and entertainment: recommending products, films, or music based on patterns in user activity.
- Agriculture: detecting crop disease from images and estimating irrigation needs from sensor data.
- Cybersecurity: identifying suspicious login patterns, malware indicators, or unusual network activity.
- Education: identifying topics where learners may need additional practice and supporting personalised learning tools.

Conclusion

Machine Learning allows computers to learn useful patterns from examples and apply them to new situations. Understanding datasets, features, labels, and the difference between training and testing data is essential for building reliable models. Supervised learning uses known labels, while unsupervised learning discovers structure in unlabelled data. Classification predicts categories, whereas regression predicts numerical values. When applied carefully and evaluated on unseen data, ML can support useful solutions in many areas of everyday life.